RAJALAKSHMI ENGINEERING COLLEGE

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T2

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|---|---|
| **Started on** | Monday, 5 October 2026, 8:46 AM |
| **State** | Finished |
| **Completed on** | Monday, 5 October 2026, 9:07 AM |
| **Time taken** | 21 mins 19 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Problem statement:

Find the length of the Longest Non-decreasing Subsequence in a given Sequence.

Eg:

Input:9

Sequence:[-1,3,4,5,2,2,2,2,3]

the subsequence is [-1,2,2,2,2,3]

Output:6

**Answer:** (penalty regime: 0 %)

```c
#include<stdio.h>

int main(){
    int n;
    scanf("%d",&n);
    int a[n];


    int dp[n];
    for(int i=0;i<n;i++){
        scanf("%d",&a[i]);
        dp[i]=1;
    }
    for(int i=0;i<n;i++){
        for(int j=0;j<i;j++){
            if(a[j]<=a[i]){
                if(dp[i]<dp[j]+1){
                    dp[i]=dp[j]+1;
                }
            }
        }
    }
    int max=dp[0];
    for(int i=1;i<n;i++){
            if(dp[i]>max){
                max=dp[i];
            }
        }

    printf("%d",max);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 9<br>-1 3 4 5 2 2 2 2 3 | 6 | 6 | ✔ |
| ✔ | 7<br>1 2 2 4 5 7 6 | 6 | 6 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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